JAYANT RATHI

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Education

Master of Science in Robotics, Northeastern University — **Boston, MA (expected Dec 2026)**

- Courses include Machine Learning and Pattern Recognition, Robotic Sensing and Navigation, Robotic Mechanics and Control, Control Systems Engineering, Reinforcement Learning and Sequential Decision Making, Electric Drives, Electric Vehicle Powertrains

Bachelor of Science in Mechatronics Engineering, Assumption University — **Bangkok, Thailand (Graduated Oct 2023)**

- Awarded Rector's Certificate of Honors for High GPA (above 3.85) for 2 consecutive years
- President, Debate Club (2019-2023) - led training, nationwide tournament operations, and public speaking development

Publication

- Published "AI in Home Security" and presented at the international EECON 46 conference (2023) as the only student in the Industry AI category, representing the ABAC engineering faculty

Technical Skills

- **Programming & Tools:** Python, C++, C, SQL, VHDL, Verilog; Linux, Docker, Git/GitLab, Conda, AWS, Firebase
- **Robotics, Control & Embedded Systems:** ROS2/ROS (Nav2, TF2), SLAM & LiDAR Odometry (LeGO-LOAM, VLP-16), Robotic Manipulation, Kinematics/Dynamics & Control (PID, LQR), State Estimation & Sensor Fusion (LiDAR/IMU/GPS), Mapping & Localization, Autonomous Navigation, Microcontrollers (ESP32, MCUs), UART/I2C/SPI, PCB Design, Sensors & Actuators
- **AI & ML:** PyTorch, TensorFlow/Keras, NumPy/Pandas; Deep Learning (CNNs), Scikit-learn, Computer Vision (OpenCV), Semantic Segmentation & Anomaly Detection, Reinforcement Learning (PPO, DQN variants, SAC, HER), Imitation Learning (ACT), Audio Classification; Hugging Face LeRobot, Stable-Baselines3, Gymnasium, MuJoCo
- **Simulation & Engineering Tools:** MATLAB/Simulink, SolidWorks, AutoCAD, Power BI, 3D CAD Engineering Design

Work Experience

Process Engineering Intern, Mercedes-Benz Thailand — **Bangkok, Thailand (Apr 2023 - Sep 2023)**

- Designed and deployed a part error-detection and replacement system, cutting a 3+ hour/day, two-person oversight process to ~45 minutes/week, then scaled the system from its Thailand pilot to Malaysia and Vietnam operations
- Built custom programs in Python, Java, and VBA for inventory/logistics management, automated shipping tracking, and data validation leading to improved accuracy and part traceability across manufacturing departments
- Acted as a go-to technical troubleshooter across plant-wide engineering initiatives, resolving system failures and optimizing data pipelines to prevent bottlenecks across the manufacturing, logistics and battery production departments
- Created real-time Power BI dashboards to improve logistics KPI visibility and help accelerate decision making for operation leadership

AI Engineer Intern, Baksters Thailand (Angel-Backed AI Startup) — **Bangkok, Thailand (Sep 2021 - Jan 2022)**

- As part of the core perception & sensor AI team, built a sound-based activity/presence and hazard detection system end to end
- Reached 97% accuracy detecting hazardous events (fires, falls, screams) by self-teaching the audio-ML stack and moving from Keras/TensorFlow classifiers to a transfer-learning approach (pretrained YAMNet embeddings + SVM)
- Led technical demos and client presentations for major SE Asian telecom clients including True Corp. and Advanced Info Service

Key Projects

Voice Controlled Robotic Arm — **Boston, MA (2026)**

- Built a voice controlled robotic manipulation system on a SO-101 arm, fine-tuning SmolVLA (a 450M parameter VLA model) on 200 self-collected teleoperated demonstrations across 4 objects for real time, on device pick and place, added a modular perception layer (OWLv2 open vocabulary detector plus target masking) that raised multi object selection from 25% to 85% with no policy retraining
- Diagnosed and fixed a train vs deploy mismatch behind grasp failures with a facing pose dataset, raising off angle success from 0/20 to 17/20 and to 80-90% across the workspace, built in PyTorch and Hugging Face LeRobot, models and datasets published on the Hub

Road Hazard Corner-Case Detection for Autonomous Driving — **Boston, MA (2026)**

- Achieved 0.69 AUPR on the Lost & Found benchmark training a DeepLabV3-ResNet50 anomaly-segmentation network from scratch (outlier-exposure learning, Cityscapes + CODA), then raised AUPR to 0.89, beating a larger pretrained Swin-B baseline (0.78), via a geometric depth-gating filter (RANSAC road-plane fitting + per-pixel height) that suppressed false positives by 60% at 95% recall

Reinforcement Learning Algorithms for Atari & Custom Game Environments — **Boston, MA (2025)**

- Developed and trained 10 custom deep reinforcement learning agents (PPO-Dueling-DQN, Quantile DQN, and Attention-DQN, etc.) across Atari games and a custom-built Cubefield game environment to benchmark performance and decision making
- Self-taught RL Concepts (e.g., policy gradients, actor-critic methods, quantile regression) and libraries such as PyTorch, Gym, and NumPy for agent design, model development and testing

Environmental 3D Mapping for Autonomous Vehicle — **Boston, MA (2024)**

- Built a real-time 6-DoF LiDAR-SLAM pipeline (LeGO-LOAM + GTSAM, ROS Noetic) for NEU's autonomous vehicle (NUance), collecting VLP-16 LiDAR, 9-axis IMU, and GPS data on driving runs and benchmarking it against LIO-SAM on the accuracy-vs-efficiency tradeoff
- Achieved ICP loop closure on 150 GB of real driving data, correcting trajectory drift and validating the path against GPS ground truth

WiFi based Fall Detection System Research — **Bangkok, Thailand (2023)**

- Designed a privacy-preserving fall-detection pipeline from WiFi Channel State Information (CSI), using a dual-ESP32 (AP/Station) setup to continuously collect fine-grained per-subcarrier RF data for motion/fall detection
- Built and labeled a dataset of 8,880 WiFi recordings (fall vs. non-fall), then trained supervised ML classifiers with feature extraction, evaluating robustness under real-world signal noise and multipath interference

AI Home Security Robot, Senior Project — **Bangkok, Thailand (2023)**

- Built a multimodal home security robot integrating perception models for facial recognition, danger sound detection, motion detection, and object detection as well as a line tracking movement system, all optimized to run simultaneously from a Raspberry Pi and Arduino
- Deployed the system live, connecting it to a Firebase cloud alert pipeline that pushed real-time notifications through LINE